Build: Dynamic Checkbox with Select All

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Problem: Display checkboxes with a Select All button. Show selected count and names. Disable the button when all are selected.

What This Tests: Array state management, derived state, controlled checkboxes.

jsx

JSX Copy

```jsx
function CheckboxGroup() {
  const [items, setItems] = useState([
    { id: 1, label: 'React', checked: false },
    { id: 2, label: 'Vue', checked: false },
    { id: 3, label: 'Angular', checked: false },
    { id: 4, label: 'Svelte', checked: false },
  ]);

  const allChecked = items.every(item => item.checked);
  const selectedItems = items.filter(item => item.checked);

  const toggleItem = (id) => {
    setItems(prev => prev.map(item =>
      item.id === id ? { ...item, checked: !item.checked } : item
    ));
  };

  const toggleAll = () => {
    const newState = !allChecked;
    setItems(prev => prev.map(item => ({ ...item, checked: newState })));
  };

  return (
    <div>
```

```
          {item.label}
        </label>
      ))}
      <button onClick={toggleAll} disabled={allChecked}>
        {allChecked ? 'All Selected' : 'Select All'}
      </button>
      <p>Selected ({selectedItems.length}): {selectedItems.map(i => i.labe
    </div>
  );
}
```

Key Concepts: Derived state (allChecked, selectedItems computed from items, not stored separately), immutable array updates with `.map()`, controlled checkboxes.

QUICK CHECK — no marks, just for you.

1. In the CheckboxGroup solution, how are allChecked and selectedItems handled?

- ○ They are stored in refs so that updating them does not trigger a re-render
- ○ They are recomputed from the items array on every render, not kept in their own state
- ○ They are stored in their own useState hooks and kept in sync inside toggleItem and toggleAll
- ○ They are computed once on mount and memoized for the lifetime of the component

Check answer

2. toggleItem maps over the previous items and returns `item.id === id ? { ...item, checked: !item.checked } : item`. After toggling the item with id 2, what is true of the resulting array?

- ○ It is a brand-new array in which only the id 2 element is a new object; the other three elements are the same object references as before

It is a brand-new array containing only the elements whose checked value changed

- It is a brand-new array in which all four elements are freshly created objects
- It is the same array as before, with the id 2 element mutated in place

Check answer

3. toggleAll computes `const newState = !allChecked` and sets every item's checked to newState, but the button is rendered with `disabled={allChecked}`. What is the practical consequence?

- The deselect-all branch of toggleAll is unreachable by clicking, because the button is disabled exactly when allChecked is true
- The button stays clickable but toggleAll becomes a no-op, leaving every item checked
- The button throws an error when clicked while all items are checked, because newState is false
- Clicking the button once all items are checked will clear every checkbox, since newState becomes false

Check answer

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Hand-picked references for this lesson.